

LLM2CLIP: Powerful Language Model Unlocks Richer Visual Representation^[1]

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[1] Weiquan Huang, Aoqi Wu, Yifan Yang, Xufang Luo, Yuqing Yang, Usman Naseem, Chunyu Wang, Qi Dai, Xiyang Dai, Dongdong Chen, Chong Luo, Lili Qiu, and Liang Hu. 2026. LLM2CLIP: Powerful Language Model Unlocks Richer Visual Representation. In Proceedings of the Conference on Artificial Intelligence. AAAI Press, 5131–5139.

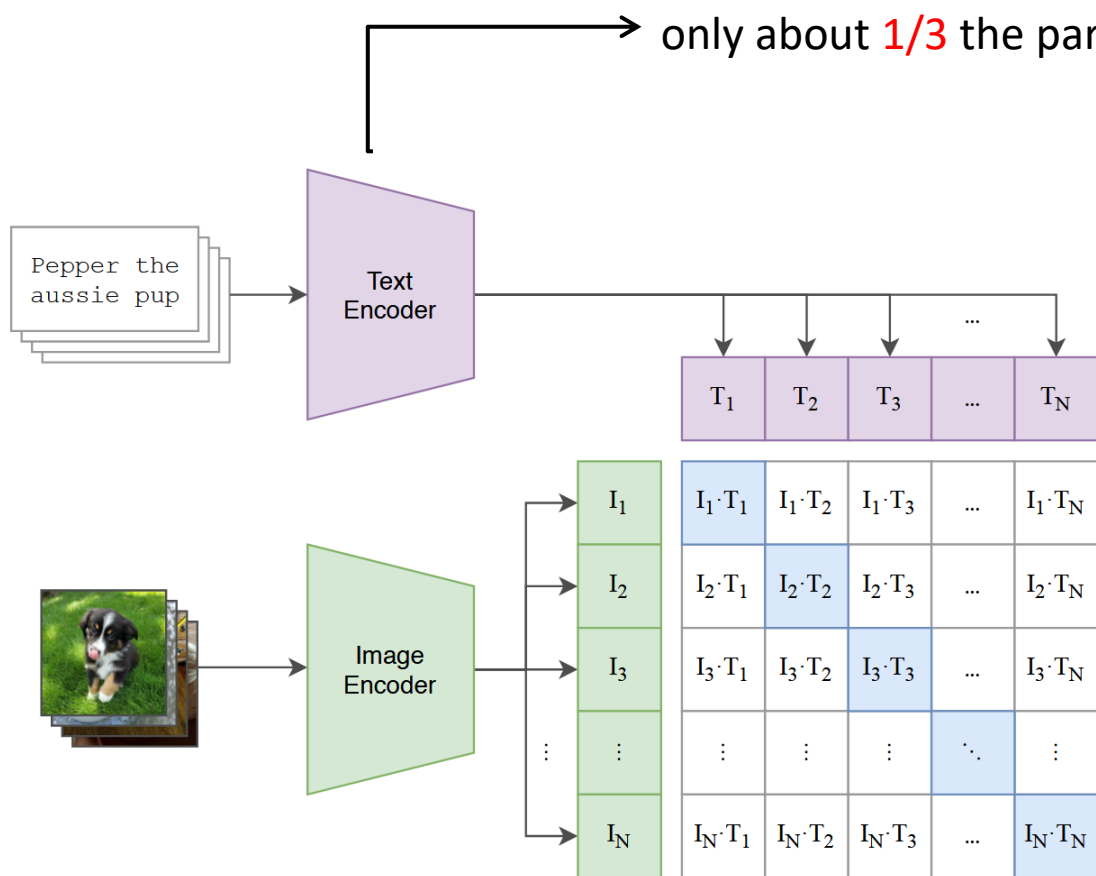


Research Questions

- Does LLM2CLIP improve image-text **alignment** compared with CLIP?
- How does **adaptor** design affect performance?
- Is LLM2CLIP more **robust** to text perturbations?



Background: Why CLIP needs LLM?



LLM has stronger language **understanding** and **open-world** knowledge

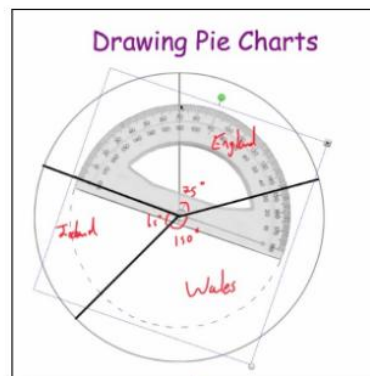
Figure 1. Contrastive pre-training of CLIP^[2]



Method Overview: Two-stage Framework

Dense Caption

"The image depicts a pie chart under construction using a protractor. The title "Drawing Pie Charts" is displayed at the top in purple text. The pie chart shows three segments labeled "England," "Ireland," and "Wales," with corresponding angle measurements in degrees..."

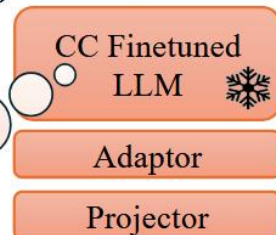
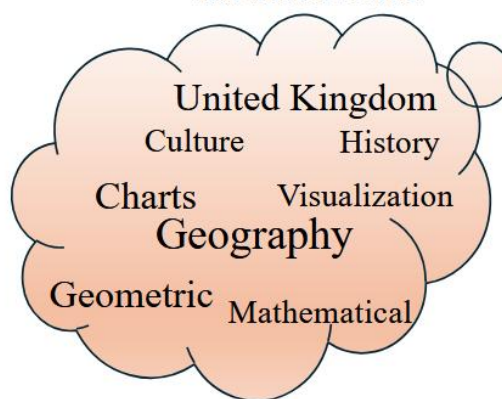


Stage 1: Convert the LLM into a caption embedding model

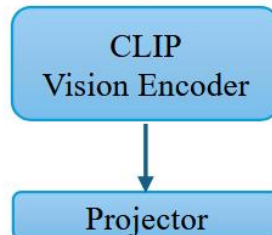
- LoRA fine-tuning
- Average pooling
- Bidirectional attention
- Supervised SimCSE

Openworld Knowledge

From LLM



Language Feature



Visual Feature

CLIP Loss

Stage 2: Replace CLIP text encoder with the fine-tuned LLM

- Freeze the LLM
- Add a lightweight adaptor
- Fine-tune the **vision encoder** and **adaptor**

Figure 2. Framework of LLM2CLIP



Experiment 1: CLIP vs. LLM2CLIP

- **Goal**
 - Compare the image-text retrieval performance of **CLIP** and **LLM2CLIP** on the Flickr dataset.
- **Dataset & Task**
 - We use the **Flickr 1K test set** for zero-shot image-text retrieval. The task has two retrieval directions:
 - **I2T: Image → Text**
Given an image, retrieve the most relevant caption.
 - **T2I: Text → Image**
Given a caption, retrieve the most relevant image.

Method

CLIP uses a vision encoder and a CLIP text encoder to map images and captions into the same embedding space.

LLM2CLIP replaces the original CLIP text encoder with a caption-contrastive fine-tuned LLM.

The LLM output is passed through a trainable adapter, which maps the LLM text embedding into a CLIP-compatible feature space.

Model	I2T	T2I
CLIP ViT-B/16	81.7	63.5
CLIP ViT-B/16 + LLM2CLIP	90.3	81.2



Experiment 2: Adaptor Modification

Model:

- a) OpenAI CLIP ViT-B/16 (encodes images into 512-d visual features for alignment with text)
- b) LLM2Vec + Llama-3-8B (encodes text into 4096-d semantic features as adaptor input)

Dataset: Flickr30k(5000/1000, every sample have 5 captions)

Structure :

- a) Default: $L2N \rightarrow 4 \times (LN \rightarrow \text{Linear} \rightarrow \text{GELU} \rightarrow \text{Linear}) + \text{Res} \rightarrow LN \rightarrow \text{Linear}$
- b) Ours: $L2N \rightarrow LN \rightarrow \text{Linear}$ (Only the text adaptor was simplified from a 4-layer MLP to LayerNorm + a single Linear layer)



Experiment 2: Adaptor Modification

T2I:Text-to-Image

I2T:Image-to-Text

Under small-scale training (5,000 images, 5 epochs), the lighter linear adaptor converges more easily, while the default multi-layer MLP is more prone to overfitting or optimization difficulty.

Task	Default Adaptor	Ours
T2I R@1	25.7%	36.0%
T2I R@5	57.4%	68.1%
T2I R@10	71.5%	80.2%
I2T R@1	52.2%	59.2%
I2T R@5	81.7%	86.7%
I2T R@10	89.9%	93.6%



Experiment 3: Text Perturbation Robustness

Model:

- a) LLM2CLIP: LM2CLIP-Llama-3-8B-Instruct-CC-Finetuned + LLM2CLIP-Openai-L-14-336
- b) CLIP: openai/clip-vit-large-patch14-336

Dataset: MSCOCO 2014 5k

Perturbation:

- a) Semantic distractor
- b) Object swap
- c) Color/spatial swap

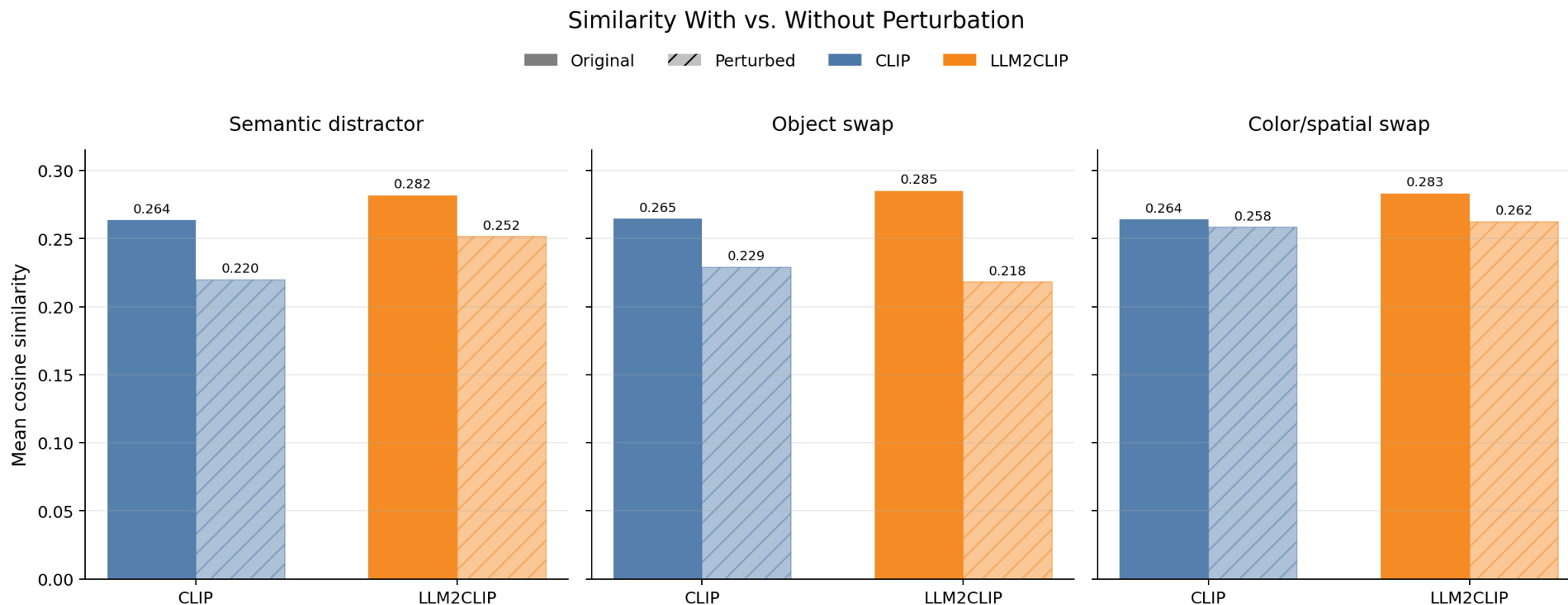
$$s_{\text{original}} = \cos(\text{image_emb}, \text{text_emb}_{\text{original_caption}})$$

$$s_{\text{corrupted}} = \cos(\text{image_emb}, \text{text_emb}_{\text{corrupted_caption}})$$

$$\delta = s_{\text{original}} - s_{\text{corrupted}}$$



Experiment 3: Text Perturbation Robustness

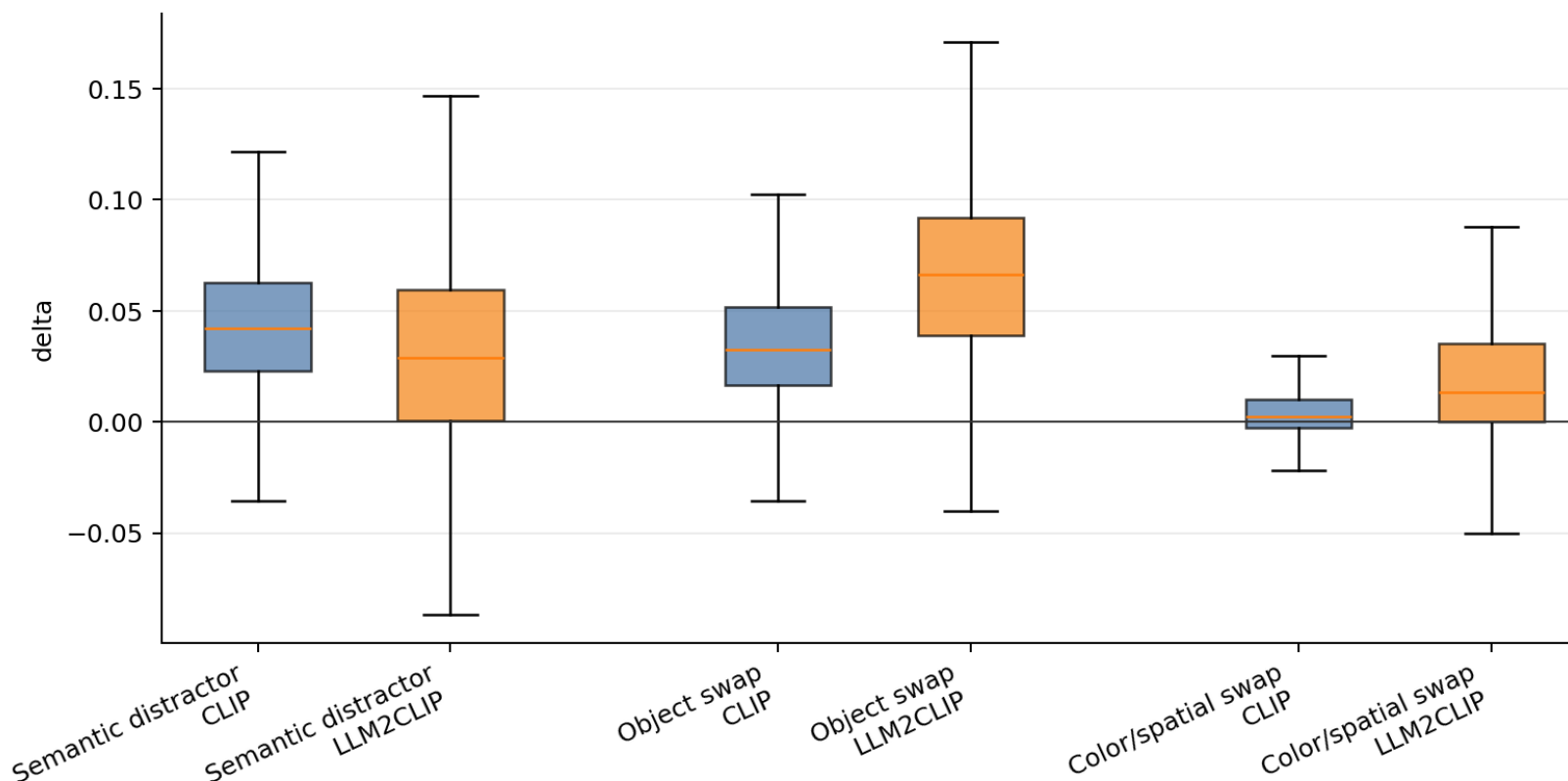


1. LLM2CLIP achieves **higher** original image-text similarity than CLIP
2. After the perturbation, all similarities **decreased**
3. Color/spatial swaps have a **smaller** impact on both models



Experiment 3: Text Perturbation Robustness

Delta Distribution by Perturbation



1. Under semantic distractors, LLM2CLIP shows a **smaller** similarity drop, suggesting better **tolerance** to **irrelevant** semantic noise
2. Under object/color/spatial swaps, LLM2CLIP shows a **larger** similarity drop, indicating stronger **sensitivity** to **fine-grained** semantic mismatch



Summary

- LLM2CLIP improves CLIP by strengthening the **text encoder**
- The adaptor enables efficient alignment with the **vision encoder**
- LLM2CLIP is **tolerant** to irrelevant noise but **sensitive** to meaningful visual-semantic changes

A series of thin, light blue wavy lines that flow from the left side of the frame, under the text, and extend towards the right side, creating a sense of movement and elegance.

Thanks For Watching!